FORDHAM
THE JESUIT UNIVERSITY OF NEW YORK

Gabelli School of Business

Lecture V – Regression Tree and ROC

Machine Learning for Business

Michael Deamer

Entropy

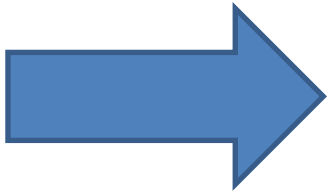
$$E = -\sum_{x=1}^{n} p_x \bullet \log_2 p_x$$

$$E = -(p1 \log_2 p1 + p2 \log_2 p2)$$

Continuous Fields – Lawn Mower

Step 1: Order the field

Income	Lot_Size	Ownership
60.0	18.4	owner
85.5	16.8	owner
64.8	21.6	owner
61.5	20.8	owner
87.0	23.6	owner
110.1	19.2	owner
108.0	17.6	owner
82.8	22.4	owner
69.0	20.0	owner
93.0	20.8	owner
51.0	22.0	owner
81.0	20.0	owner
75.0	19.6	non-owner
52.8	20.8	non-owner
64.8	17.2	non-owner
43.2	20.4	non-owner
84.0	17.6	non-owner
49.2	17.6	non-owner
59.4	16.0	non-owner
66.0	18.4	non-owner
47.4	16.4	non-owner
33.0	18.8	non-owner
51.0	14.0	non-owner
63.0	14.8	non-owner

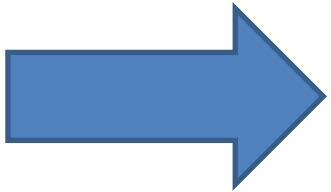


Income	LotSize	Owner
51	14	No
63	14.8	No
59.4	16	No
47.4	16.4	No
85.5	16.8	Yes
64.8	17.2	No
108	17.6	Yes
84	17.6	No
49.25	17.6	No
60	18.4	Yes
66	18.4	No
33	18.8	No
110.1	19.2	Yes
75	19.6	No
69	20	Yes
81	20	No
43.2	20.4	No
61.5	20.8	Yes
93	20.8	No
52.8	20.8	No
64.8	21.6	Yes
51	22	No
82.8	22.4	Yes
87	23.6	Yes

Continuous Fields – Lawn Mower

Step 2: Calculate midpoints

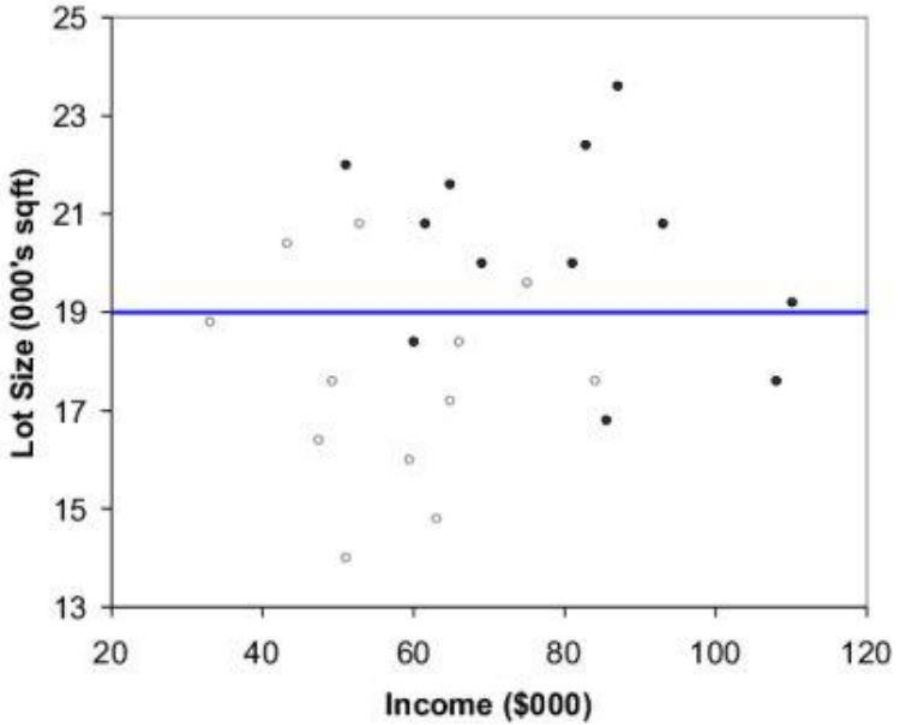
Income	LotSize	Owner
51	14	No
63	14.8	No
59.4	16	No
47.4	16.4	No
85.5	16.8	Yes
64.8	17.2	No
108	17.6	Yes
84	17.6	No
49.25	17.6	No
60	18.4	Yes
66	18.4	No
33	18.8	No
110.1	19.2	Yes
75	19.6	No
69	20	Yes
81	20	No
43.2	20.4	No
61.5	20.8	Yes
93	20.8	No
52.8	20.8	No
64.8	21.6	Yes
51	22	No
82.8	22.4	Yes
87	23.6	Yes



Income	LotSize	Owner	
51	14	No	14.4
63	14.8	No	15.4
59.4	16	No	16.2
47.4	16.4	No	16.6
85.5	16.8	Yes	17
64.8	17.2	No	17.4
108	17.6	Yes	17.6
84	17.6	No	17.6
49.25	17.6	No	18
60	18.4	Yes	18.4
66	18.4	No	18.6
33	18.8	No	19
110.1	19.2	Yes	19.4
75	19.6	No	19.8
69	20	Yes	20
81	20	No	20.2
43.2	20.4	No	20.6
61.5	20.8	Yes	20.8
93	20.8	No	20.8
52.8	20.8	No	21.2
64.8	21.6	Yes	21.8
51	22	No	22.2
82.8	22.4	Yes	23
87	23.6	Yes	11.8

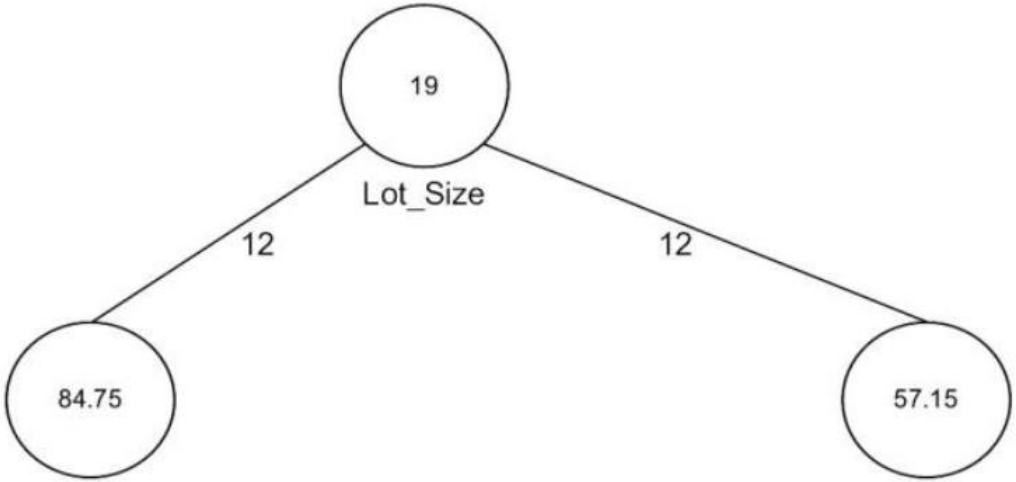
Continuous Fields – Lawn Mower

Step 3: Split on each midpoint, evaluate, and repeat



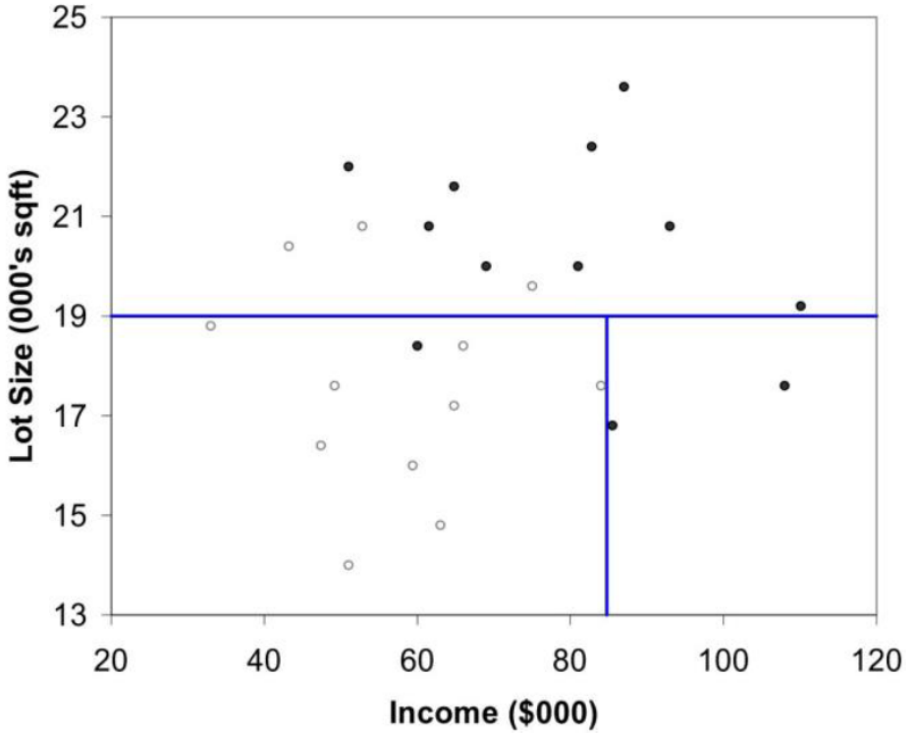
Continuous Fields – Lawn Mower

Step 4: Branch tree at optimal split point



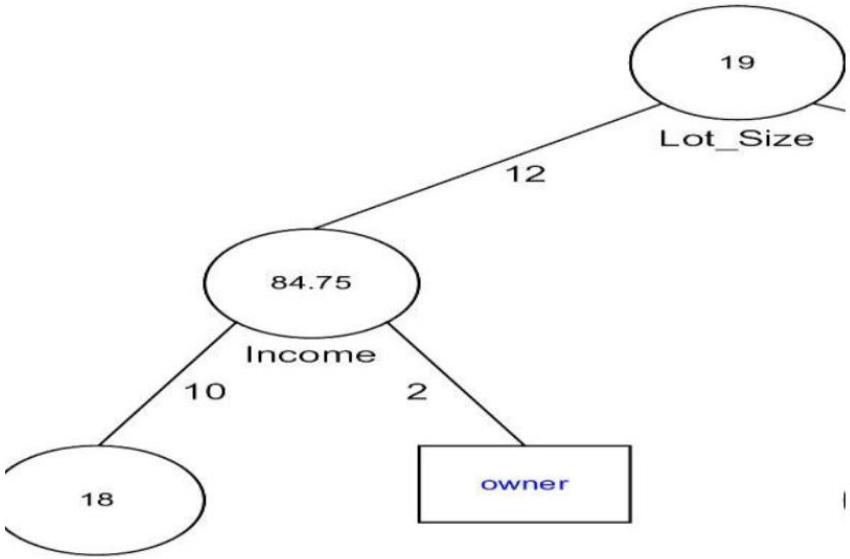
Continuous Fields – Lawn Mower

Step 5: Repeat for other fields



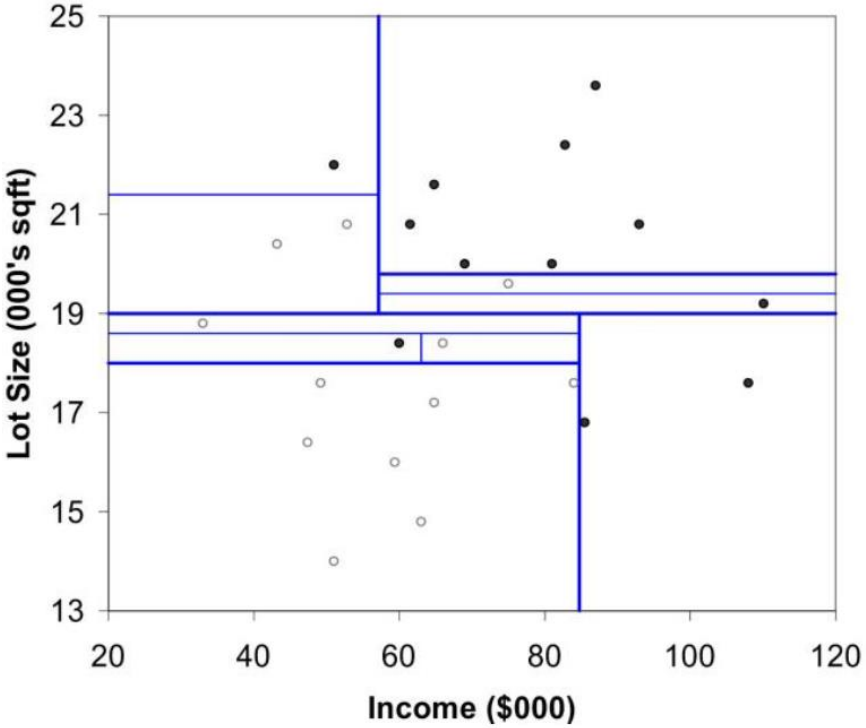
Continuous Fields – Lawn Mower

Step 4: Branch tree at optimal split point



Continuous Fields – Lawn Mower

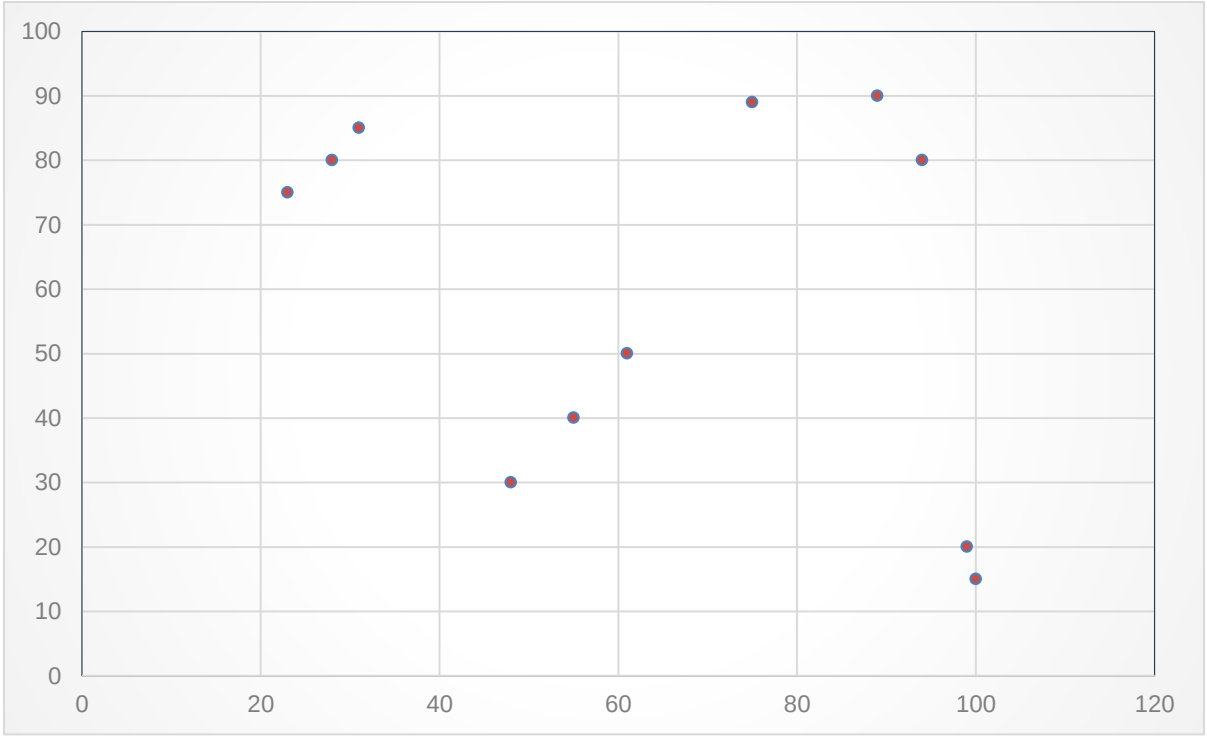
Result



Predicting Continuous Target Field

Games Completed

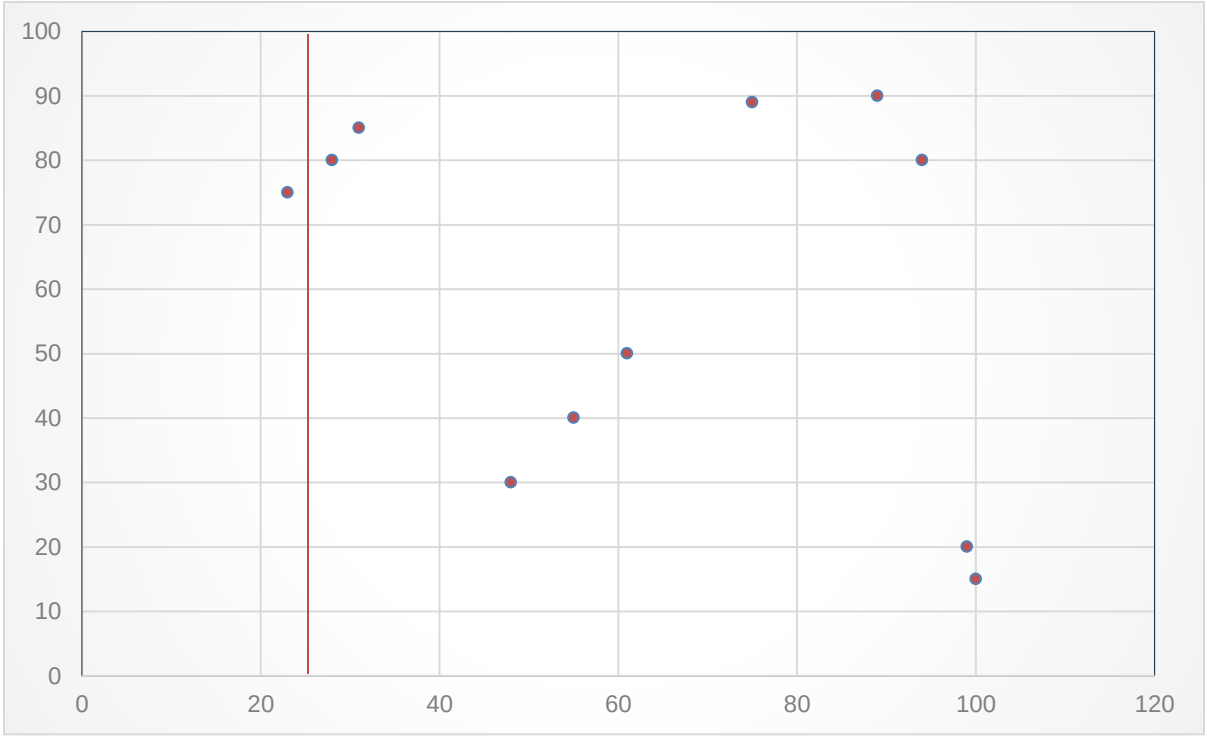
Temp	Games
23	75%
28	80%
31	85%
48	30%
55	40%
61	50%
75	89%
89	90%
94	80%
99	20%
100	15%



Splitting at 25.5

Games Completed

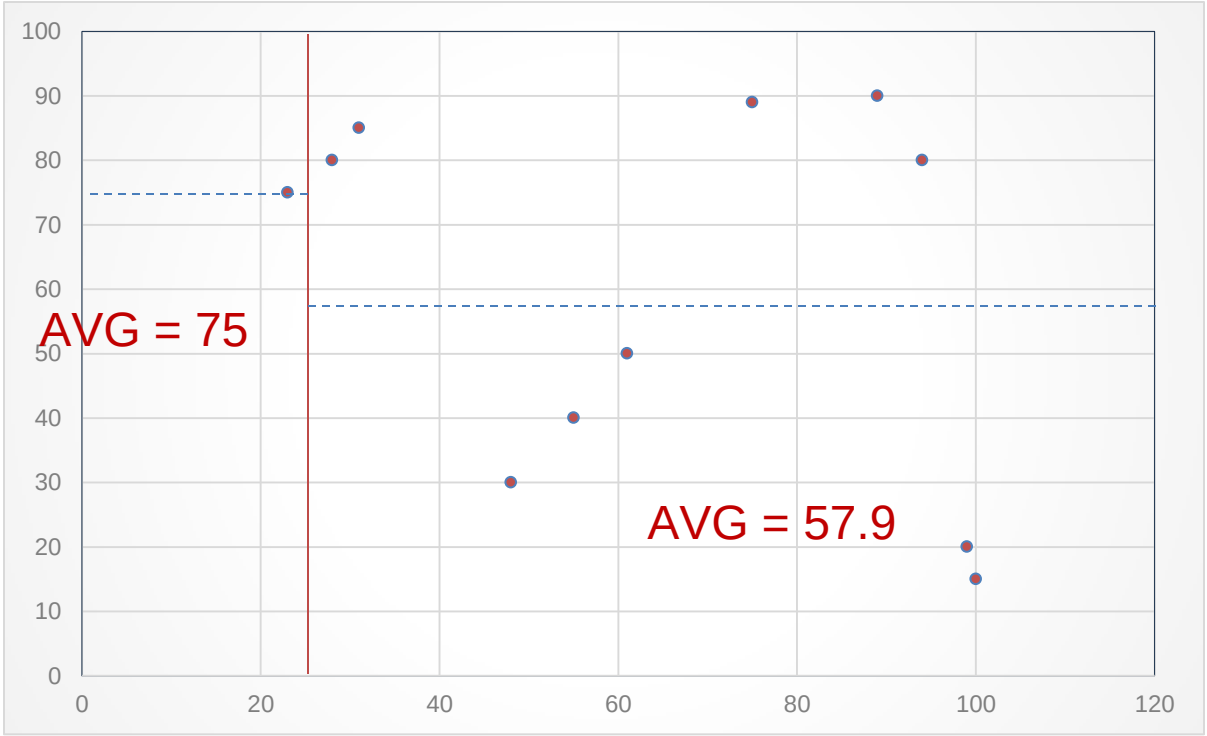
Temp	Games
23	75%
28	80%
31	85%
48	30%
55	40%
61	50%
75	89%
89	90%
94	80%
99	20%
100	15%



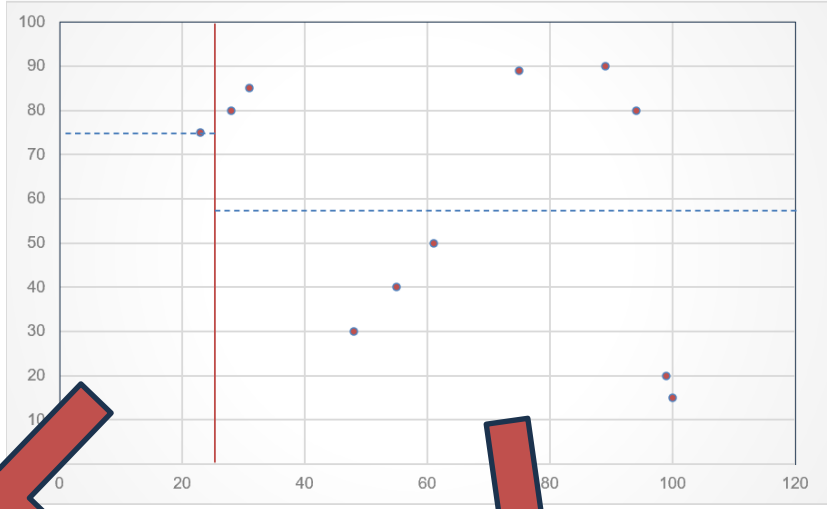
Find Average of Each Partition

Games Completed

Temp	Games
23	75%
28	80%
31	85%
48	30%
55	40%
61	50%
75	89%
89	90%
94	80%
99	20%
100	15%



Find Residual Sum or Squares



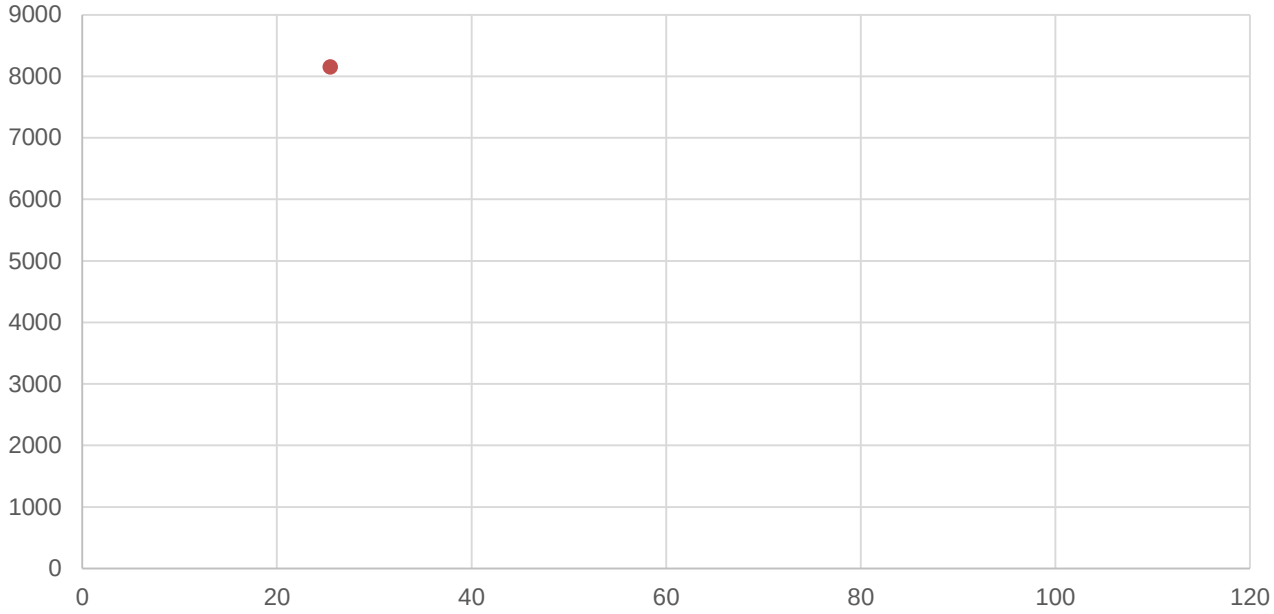
$$RSS = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$(75 – 75)^2$ +

$(80 – 58)^2 + (85 - 58)^2 + (30 - 58)^2 + (40 - 58)^2 + (50 - 58)^2 +$
$(89 - 58)^2 + (90 - 58)^2 + (80 – 58)^2 + (20 - 58)^2 + (10 - 58)^2$

=8147

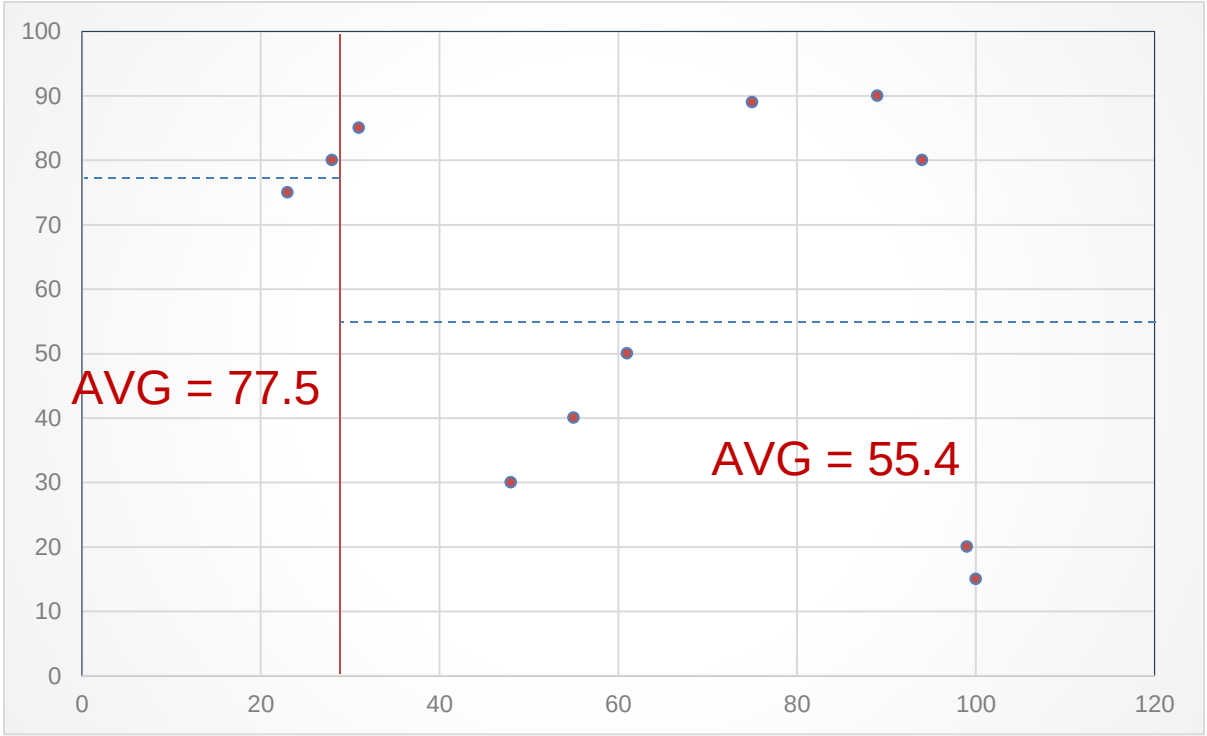
Residuals


9000
8000
7000
6000
5000
4000
3000
2000
1000
0
0
20
40
60
80
100
120

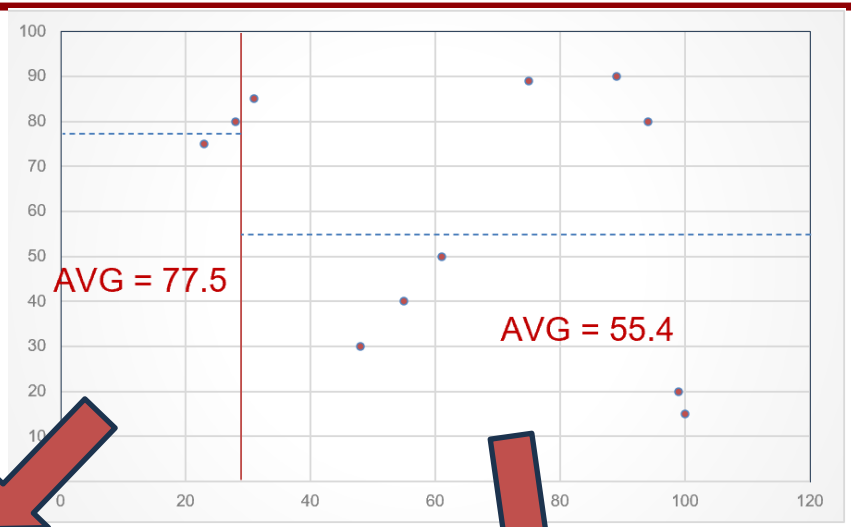
Predicting Continuous Target Field

Games Completed

Temp	Games
23	75%
28	80%
31	85%
48	30%
55	40%
61	50%
75	89%
89	90%
94	80%
99	20%
100	15%



Find Residual Sum or Squares



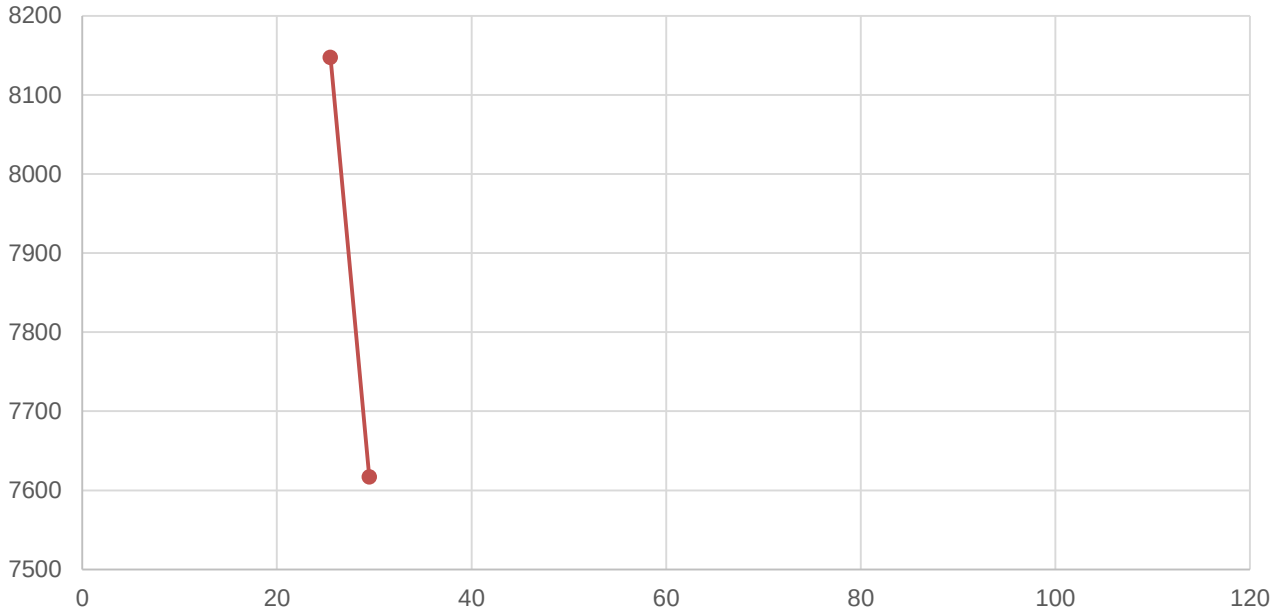
$$RSS = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$(75 – 77.5)^2 + (80 – 77.5)^2 +$

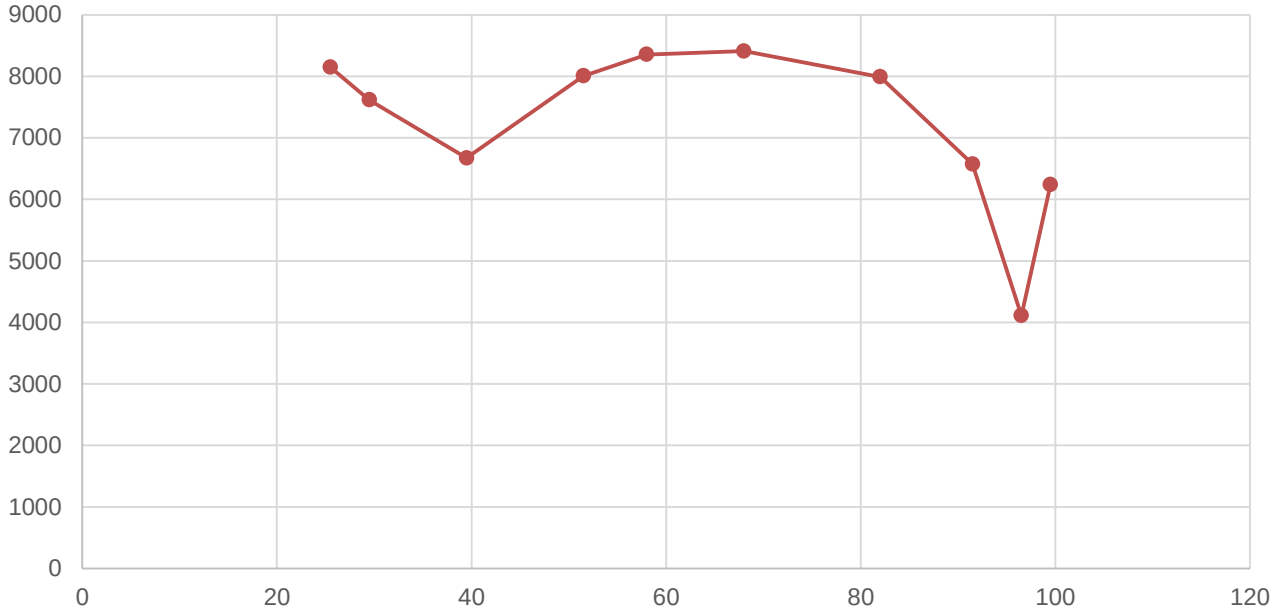
$(80 – 55.4)^2 + (85 - 55.4)^2 + (30 - 55.4)^2 + (40 - 55.4)^2 + (50 - 55.4)^2 + (89 - 55.4)^2 + (90 - 55.4)^2 + (80 – 55.4)^2 + (20 - 55.4)^2 + (10 - 55.4)^2$

=7616

Residuals

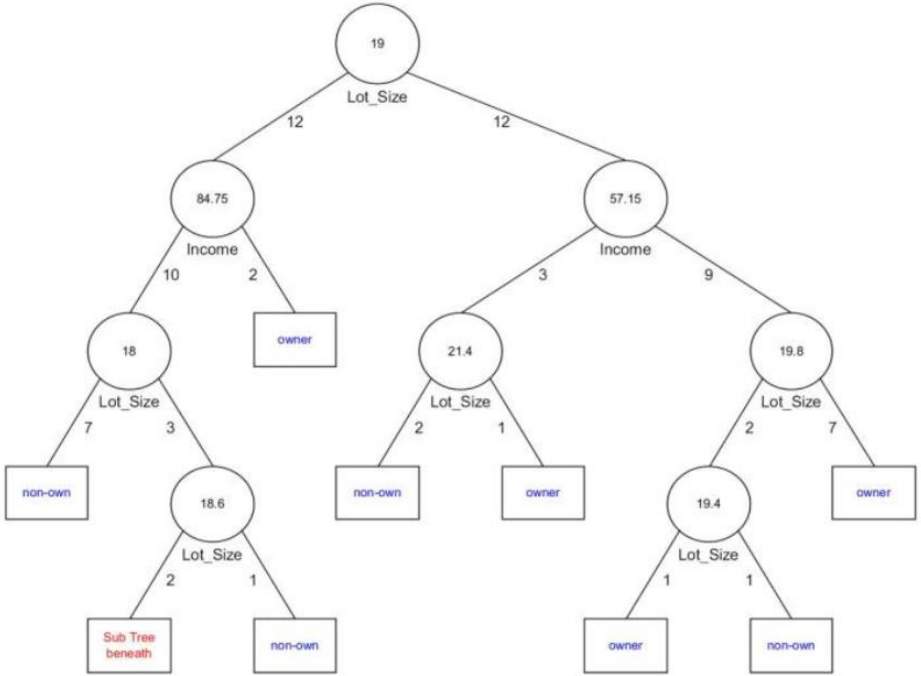


Residuals


9000
8000
7000
6000
5000
4000
3000
2000
1000
0
0
20
40
60
80
100
120

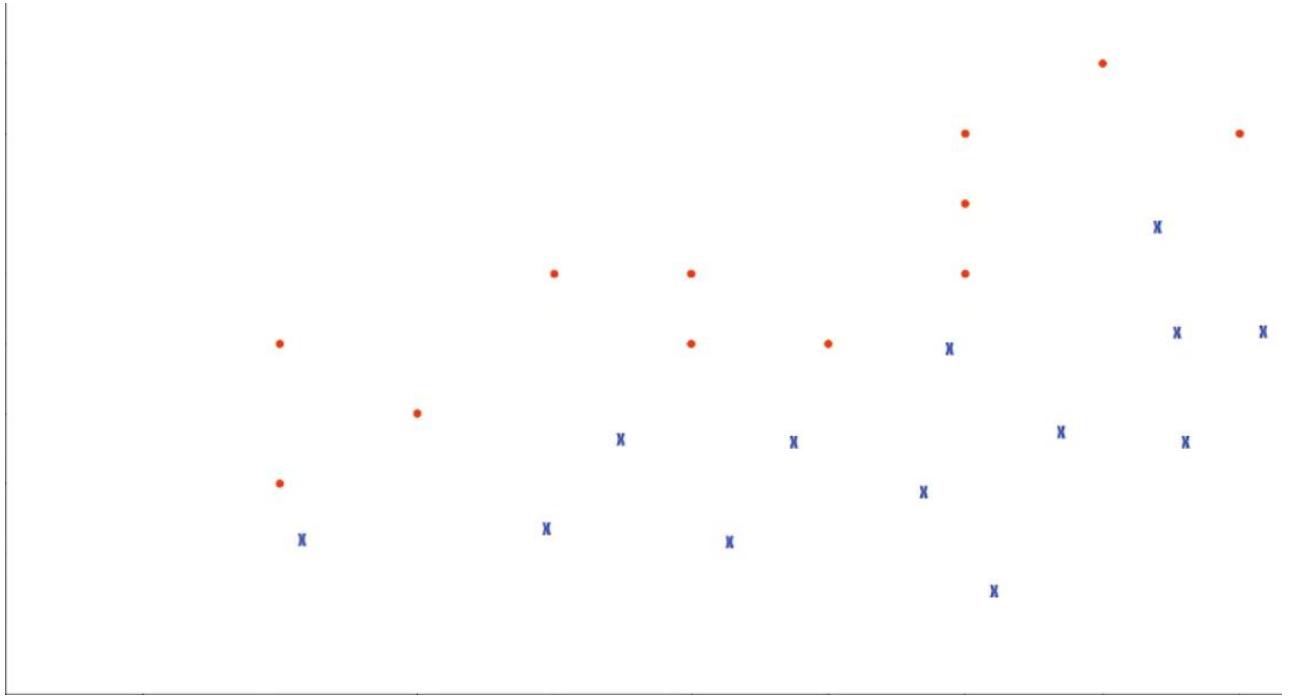
Continuous Fields – Lawn Mower

Result Tree



Limitation

Interaction between variables



To Do

- ❑ Lab IV and Homework IV
- ❑ Prepare for Exam I
- ❑ Connect with group via email and schedule regular meetings

Bibliography

Tan, Steinbach, Karpatne, and Kumar. Introduction to Data Mining, second edition. Pearson, Hudson, NY, 2019.
Han, Kamber, Pei. Data Mining Concepts and Techniques, second edition. University of Illinois at Urbana-Champaign. Elsevier, San Francisco, CA, 2006.